

PREPARED BY THEMISIQ COMPLIANCE INC.

Climate Risk & Scenario Analysis Report

IFRS S2 — single (financial) materiality screening for climate-related risks and opportunities.

Legal entity	Acme Industries
Reporting period	FY2025
Primary sector	Industrials & Manufacturing
Operating regions (IPCC AR6)	Mediterranean (MED), Eastern North America (ENA)
Policy jurisdictions	US (federal), EU (EU ETS), EU CBAM exposure
Scenario tested	IPCC SSP2-4.5 (~2.7°C)
Time horizon	medium term
Asset profile	inland
Model version	0.1-draft
Report date	May 27, 2026

Executive summary

This single materiality screening identifies climate-related risks and transition exposures that are likely to be material for **Acme Industries** under the IPCC SSP2-4.5 pathway over the medium term.

2

High physical risks

1

High transition risks

Key findings

Physical risk: **Extreme heat** is material (high band, score 7.2) — driven by exposure in **MED**.

Physical risk: **Water stress** is material (high band, score 6) — driven by exposure in **MED**.

Transition risk: **Technology displacement** is material (high band) under the chosen scenario and jurisdictions.

Methodology and basis

This screening combines four public, independently-sourced frameworks. No proprietary or licensed third-party classification is reproduced.

Frameworks

IPCC AR6 WGI reference regions — the 20 land regions used here are drawn from the Sixth Assessment Report Working Group I reference-region set (Iturbide et al., 2020), each with its characteristic profile of climatic impact-drivers.

IPCC climatic impact-drivers — the physical hazards (extreme heat, drought, water stress, inland flooding, coastal flooding, wildfire, storms/cyclones, cold/permafrost) follow the AR6 climatic-impact-driver categories.

TCFD transition risk categories — transition risks follow the Task Force on Climate-related Financial Disclosures classification: policy and legal, technology, market, and reputation.

Scenario pathways — both IPCC Shared Socioeconomic Pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5) and NGFS scenarios (Orderly, Disorderly, Hot House) are available; this screening uses IPCC SSP2-4.5.

Risk model

Physical risk is computed as industry sensitivity × regional hazard exposure × scenario severity × time-horizon multiplier. A risk is flagged only where industry sensitivity and regional hazard exposure intersect — preventing the common error of flagging, for example, drought for any agricultural entity regardless of where it operates.

Transition risk is computed as industry carbon exposure × jurisdictional policy intensity × scenario policy-speed × time-horizon multiplier. Transition geography (policy jurisdictions) is deliberately distinct from physical geography (IPCC regions): physical exposure depends on where assets and hazards are; transition exposure depends on which policy regimes apply.

Scenario inversion

Physical and transition risk move in opposite directions across scenarios: high-warming pathways raise physical risk and lower transition pressure; rapid- or abrupt-policy pathways raise transition risk and lower physical pressure. A resilient strategy must hold up across both ends, which is why both frameworks require testing resilience across a range rather than against a single forecast.

Scenario selection and rationale

Pathway used: IPCC SSP2-4.5 (~2.7°C).

A middle pathway (SSP2-4.5, ~2.7°C) was selected as a reasonable central case for first-pass screening. Higher- or lower-warming pathways can be tested to assess resilience across a range.

Time horizon used: medium term.

A medium-term horizon (to 2040) was selected as the default screening lens. Companies with long-lived physical assets may prefer the long-term view; near-term commitments may benefit from the short-term view.

Under both IFRS S2 and CSRD/ESRS, the choice of scenarios used and the rationale for that choice are themselves disclosable. This section documents that judgment.

Risk register

Physical risks

Industry sensitivity × regional hazard exposure × scenario × horizon. Flagged only where industry sensitivity meets real regional hazard exposure.

Hazard	Severity	Driving region	Score
Extreme heat	HIGH	Mediterranean (MED)	7.2
Water stress	HIGH	Mediterranean (MED)	6.0
Inland flooding	MED	Eastern North America (ENA)	4.0

Transition risks

Industry carbon exposure × jurisdictional policy intensity × scenario policy-speed × horizon.

Driver	Severity
Carbon pricing / policy	MED
Market & demand shift	MED
Technology displacement	HIGH
Reputation	MED